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Predicting financial distress in emerging markets: the case of Indian small enterprises

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ABSTRACT: The present study examines the factors leading to financial distress among small enterprises in India. The CMIE Database was used to extract data from 2008 to 2022. A total of 946 companies' data has been extracted; however, due to the missing data, 547 companies' data were used for the analysis. The Springate S-Score is used to classify firms as healthy or distressed, and the logit regression is used to predict the financial health of small firms by using financial ratios, firm-specific, and macroeconomic variables. The findings revealed that interest expense ratio, current ratio, leverage ratio, total asset turnover ratio, size of the company, book value of equity, GDP growth, inflation, and economic policy uncertainty have significance in predicting a small company's financial health position. Model 1 has an accuracy rate of 92.7 percent, and Model 2 corroborated the robustness of the findings. The present study addresses an essential requirement for understanding the economic health of small companies in emerging economies like India. At the same time, it provides valuable insights for small companies, investors, financial analysts, and policymakers to improve prediction accuracy and develop strategies to mitigate the risk.

Impact statement: This study addresses a critical gap in financial distress prediction by focusing on small companies in emerging markets like India, which have been underrepresented in prior research. By integrating financial, firm-specific, and macroeconomic variables, the study develops highly accurate predictive models (Model 1: 92.47%, Model 2: 92.4%) that identify key drivers of financial instability, such as liquidity ratios, leverage, profitability, and external economic factors like GDP growth and inflation. The research holds significant academic and practical value. For financial institutions, the models enhance credit risk assessment for small businesses, while policymakers can leverage insights to design targeted support mechanisms. Small business owners gain actionable knowledge on financial health indicators, enabling proactive risk management. Additionally, the study underscores the importance of macroeconomic stability in safeguarding small enterprises, offering policymakers evidence-based strategies to mitigate economic volatility. This research bridges theory and practice to enhance small business resilience in emerging markets. By identifying key financial distress predictors, it equips stakeholders with actionable tools for informed decision-making and sustainable growth strategies.

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1. Introduction

A number of business entities find themselves in a fragile financial situation due to the current economic environment. Financial distress refers to businesses struggling to pay their bills, keep up with debt payments, or even make payroll (Mselmi et al., 2017). It can be caused by a lack of cash flow, poor financial management, or external factors like a recession or a global pandemic (ElBannan, 2021). Financial distress often leads to poor creditworthiness, loss of investor trust, and ultimately insolvency

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(Geng et al., 2015). In an increasingly unstable economic world, accurately predicting financial distress has become imperative for protecting businesses and sustaining market stability. Small and microbusiness entities are particularly vulnerable to economic conditions due to limited financial resources and restricted market access (Ciampi et al., 2021). Financial distress can have devastating effects for a company that lead to bankruptcy (Geng et al., 2015; Giammarino, 1989). Bankruptcy is a legal process that allows a company to restructure or discharge its debts, but it also means that the company's assets are liquidated and may result in ceasing operations (Bolton et al., 2011). Interestingly, Altman (1968) describes bankruptcy as the closest definition of financial distress. However, many firms avoid filing for bankruptcy through acquisition or privatization, and on the other hand, healthy and high-performing firms often file for bankruptcy strategically to evade tax filings and other legal suits (Geng et al., 2015). Thus, effective utilization of financial resources plays a significant role in preventing financial distress and bankruptcy (Li et al., 2020; Shen et al., 2020). This can be achieved through strategies such as monitoring financial statements, developing sustainable financial plans, minimizing costs, increasing revenue, securing additional funding, and consulting financial advisors (Jones & Wang, 2019).

The existing body of research on financial distress in India has primarily focused on larger corporations and neglected the unique challenges faced by small firms. Small-scale companies play a vital role in the economic growth of developing economies. However, these companies face numerous challenges in sustaining themselves in the market. Small firms are significantly affected by market conditions, firm-level attributes, and macroeconomic conditions (Betz et al., 2014; Habib et al., 2020; Yazdanfar & Öhman, 2020). Identifying the early indicators of financial distress can help stakeholders-including creditors, investors, and policymakers take proactive measures to mitigate risks. Therefore, this significant gap encourages analysis of financial ratios, firm-level attributes, and macroeconomic conditions that influence a small firm's financial health.

Bankruptcy has serious consequences for the company and the economy as a whole. Consequently, extensive research identifies the potential risks to avoid economic losses (Jackson & Wood, 2013; Liang et al., 2016; Pindado et al., 2008). To reduce risk, all parties involved in the company should be aware of its financial situation. Creditors or financial institutions can make decisions carefully and minimize the losses by adjusting the interest rate and minimizing the NPAs (Chen et al., 2020; Hernandez Tinoco & Wilson, 2013; Li & Faff, 2019). A strong predictive model can guide investors in making profitable decisions (Tian & Yu, 2017), while regulatory bodies can implement policy-level interventions to assist small-scale companies. As a result, the role of macroeconomic and firm-level variables in predicting bankruptcy has gained prominence (Habib et al., 2020).

Companies can improve their resilience by implementing internal risk mitigation strategies to overcome financial challenges and ensure business continuity, particularly for small enterprises (Cheraghali & Molnár, 2024). However, predicting financial distress remains challenging due to the complex interactions between economic and business factors (Opler & Titman, 1994). Existing models have not effectively assessed financial health, especially for small companies. Therefore, a comprehensive framework that incorporates financial variables, firm attributes, and macroeconomic indicators is necessary (Boritz et al., 2007; Pindado et al., 2008). This study focuses on developing a predictive model by combining financial, firm, and macroeconomic variables to address the unique risks faced by small business entities. The study aims to enhance the accuracy of predictions and provide valuable insights for financial analysts, policymakers, and business leaders. These insights will help formulate robust financial strategies and identify key factors contributing to financial distress.

2. Literature review

Predicting financial distress has become essential for stakeholders to mitigate risks associated with potential insolvencies (Li et al., 2021). Researchers have developed various models by incorporating financial ratios, firm variables, and macroeconomic indicators to enhance predictive accuracy (Chen et al., 1995). This literature review consolidates recent studies on financial distress and highlights the combined effectiveness of multiple variables.

Financial ratios are fundamental for predicting a firm's financial distress. Beaver (2010) and Li et al. (2020) found that financial ratios possess strong predictive power to identify financial distress.

Furthermore, advanced statistical methods can enhance accuracy and applicability across industries and firm sizes. Chen et al. (2020) emphasized the need for continuous refinement of statistical models that utilize financial ratios.

Market variables also play a predominant role in improving the financial distress prediction models. Especially market variables have a direct relationship with the financial conditions that can improve the model. Gilson (1989) and Liang et al. (2016) observed that the integrating market indicators with traditional financial ratios improves the model's accuracy. To further refine predictions, Pindado et al. (2008) and Haleem (2020) proposed dynamic models that use market variables to account for companies' different levels of credit risk.

Moreover, macroeconomic indicators have gained prominence in financial distress prediction. Li and Miu (2010) and Rico et al. (2021) found a significant association between macroeconomic indicators and financial distress, suggesting that these indicators are essential for regional and sector-specific predictions. Integrating macroeconomic indicators provides a broader context, helping predict distress in dynamic economic conditions across different markets (Boratyńska & Grzegorzewska, 2018; Li & Miu, 2010; Pindado et al., 2008).

Several studies proposed a collective approach that combines financial, firm, and macroeconomic variables to predict distress. Pindado et al. (2008) proved that combining all these variables not only has better predictive accuracy but also offers practical value for risk management and decision-making. Similarly, this integrated approach is supported by the findings of (Boratyńska & Grzegorzewska, 2018; Hernandez Tinoco & Wilson, 2013), who suggest that the inclusion of multiple data sources enhances the models' practical utility and applicability.

Existing literature confirms that the current models can predict distress with better accuracy. However, there is significant scope for further research to refine these models. Kou et al. (2021) and Li et al. (2020) highlighted the prominence of combining further types of financial and nonfinancial information that explore the effect of different accounting standards and the impact of management actions on bankruptcy risk. Additionally, exploring the applicability of the existing models across various sectors and under diverse economic conditions is crucial for ensuring their robustness and relevance (Beaver, 2010; El Kalak & Hudson, 2016; Kou et al., 2021).

2.1. Theory and hypothesis development

2.1.1. Cash management theory

Cash management theory refers to managing cash balances as a critical concern for every firm due to the inherent difficulty in accurately forecasting cash flows, especially inflows. The lack of perfect alignment between cash inflows and outflows can indicate a failure in the firm's cash management practices. Such an imbalance could ultimately lead to financial distress and potential business failure (Adnan Aziz & Dar, 2006). As per the theory, we propose the following hypothesis.

H1: There is a significant negative relationship between cash flow and the likelihood of financial distress in firms.

According to Hashi (1997), the liquidity and profitability theory states that a firm is considered healthy when it exhibits strong liquidity and profitability indicators. Conversely, if these indicators are weak, the firm is viewed as unhealthy and at risk of bankruptcy. High and positive levels of these two indicators suggest a lower risk of bankruptcy (Duong et al., 2024; Isayas, 2021).

2.1.2. Financial ratios

Several experiments show that financial ratios predict financial distress (Altman, 1968; Bauer & Agarwal, 2014). Accounting data can help identify financially distressed firms. The cash flow theory states that firms are considered financially strong if their operations generate sufficient cash flow (Liang et al., 2016). The liquidity ratio shows a firm's ability to meet short-term obligations with highly liquid assets. Firms with higher quick ratios are financially healthier (Boritz et al., 2007). Working capital to total assets states how much working capital depends on company assets. Lower ratios indicate better asset use

(Habib et al., 2020). Return on assets is used to measure the net profit generated by each unit of assets. This ratio indicates how efficiently a firm utilizes its assets to generate profit (Wu et al., 2010). A higher ROA indicates better efficiency. The leverage ratio shows debt-to-equity and it clarifies firm capital structures. Higher leverage ratios indicate financial risk and distress (Farooq et al., 2023; Haleem, 2020). The firm's total asset to total liability (TLTA) shows how its assets cover its liabilities (Boratyńska & Grzegorzewska, 2018; El Kalak & Hudson, 2016). The net income ratio measures the proportion of net income a company earns relative to its total revenue (Habib et al., 2020). Financial expenses help assess a firm's leverage and risk, and high expenses lower net profitability (Hernandez Tinoco & Wilson, 2013). The total asset turnover ratio assesses how effectively a company utilizes its assets to produce sales (Alifiah, 2014). As per prior studies, we propose the following hypothesis.

H2: There is a significant negative relationship between liquidity ratios and the likelihood of financial distress in firms.

H3: There is a significant negative relationship between net income ratio and the likelihood of financial distress in firms.

H4: There is a significant positive relationship between interest expenses ratio and the likelihood of financial distress in firms.

H5: There is a significant negative relationship between solvency ratio and the likelihood of financial distress in firms.

H6: There is a significant positive relationship between total assets turnover ratio and the likelihood of financial distress in firms.

H7: There is a significant negative relationship between return on assets and the likelihood of financial distress in firms.

2.1.3. *Firm variables*

Firm variables can directly assess volatility and are powerful predictors of bankruptcy risk. Firm size and book value of equity are studied as firm variables. Large firms have more market influence and growth resources, which can reduce business risk and benefit economies of scale, lowering costs and increasing profit margins (Chen et al., 2020; El Kalak & Hudson, 2016; Isayas, 2021; Liang et al., 2016). The book value of equity measures the company's net value and financial stability (Khoja et al., 2019). As per prior studies, we propose the following hypothesis.

H8: There is a significant negative relationship between size and the likelihood of financial distress in firms.

H9: There is a significant positive relationship between book value of equity and the likelihood of financial distress in firms.

2.1.4. *Macroeconomic variables*

According to Altman et al. (2017), Betz et al. (2014), Hernandez Tinoco and Wilson (2013), Yazdanfar and Öhman (2020), and Habib et al. (2020), incorporating country risk factors into models predicting firm-level financial distress significantly enhances their ability to explain outcomes. GDP growth measures national economic performance. Positive GDP growth indicates economic health (Zoričák et al., 2020). The inflation shows the average consumer price change over time. It's the key inflation indicator. Higher CPIs indicate higher inflation, which lowers purchasing power (Altman et al., 2020). Economic policy uncertainty affects investor and expansion decisions, economic growth, and business performance. High uncertainty can lead to increased market volatility and risk (Li et al., 2021; Phan et al., 2021). As per prior studies, we propose the following hypothesis.

H10: There is a significant negative relationship between GDP growth and the likelihood of financial distress in firms.

H11: There is a significant positive relationship between inflation and the likelihood of financial distress in firms.

H12: There is a significant negative relationship between uncertainty index and the likelihood of financial distress in firms.

The existing literature on predicting financial distress shows a trend toward multiple-variable models that combine financial ratios, firm variables, and macroeconomic indicators (Betz et al., 2014; Hernandez Tinoco & Wilson, 2013; Yazdanfar & Öhman, 2020). This approach not only improves the accuracy of predictions but also aligns with practical needs for risk assessment and management. We can understand and predict the financial crisis as these models continue to evolve.

3. Data and methods

The present study attentions on creating a financial distress prediction model that combines financial, firm, and macroeconomic factors in order to anticipate the financial distress of small firms in India through a quantitative study.

This study depends on secondary data sources that include financial and firm-level data from the CMIE database, macroeconomic indicators from the World Bank, and uncertainty index figures from policy uncertainty websites.

We specifically examine small-scale companies listed on the Bombay Stock Exchange (BSE). For our sample selection, we employed purposive sampling, initially collecting data from all 946 companies in the S&P BSE SmallCap index between 2008 and 2022. After filtering, we found complete data sets for 547 companies. Thus, our analysis is based on an imbalanced panel dataset of 7,457 company-year observations from these 547 companies, ensuring a robust and accurate evaluation.

The analysis is conducted using Gretl software and employs logit regression as the primary statistical method. The logit regression was used to estimate the likelihood of financial distress among the selected companies. The Springate S-Score model was employed to determine whether the companies were healthy or in distress.

3.1. Dependent variable – Springate s-score

Gordon Springate created the model in 1978; it employs stepwise discriminant analysis to assign scores to each business. Nineteen widely used financial ratios were initially incorporated into the Springate model. Nevertheless, after extensive testing, Springate ultimately selected four financial ratios to assess whether a company is healthy or distressed. According to Husein and Pambekti (2015), the Springate model serves as a tool for evaluating a company's condition and performance for stakeholders.

Table 1. Independent variables used in the study.

Variable name	Short form	Reference
Cash flow from operations to sales	CFO	Waqas and Md-Rus (2018)
Current ratio	CR	Mselmi et al. (2017)
Quick ratio	QR	Mselmi et al. (2017); Chen et al. (2020)
Net working capital	NWC	Mselmi et al. (2017)
Net income ratio	NIR	Habib et al. (2020)
Interest expense ratio	IER	Hernandez Tinoco and Wilson (2013)
Leverage ratio	LR	Waqas and Md-Rus (2018)
TLTA	TLTA	Bauer and Agarwal (2014)
Total assets turnover	TAT	Alifiah (2014)
Return on assets	ROA	Qian et al. (2022)
Size of the firm	Size	Rico et al. (2021); Waqas and Md-Rus (2018)
Book value of equity	BE	Supriyanto and Darmawan (2018)
GDP growth	GDPG	Li et al. (2021)
Inflation	IN	Pham Vo Ninh et al. (2018)
Uncertainty index	UI	Phan et al. (2021)

Additionally, the Springate model is recognized for its ability to predict future bankruptcy, functioning effectively as an early warning system for potential financial distress.

The Springate S-Score formula is calculated as follows: ' $S = 1.03A + 3.07B + 0.66C + 0.4D$ '. Here, A stands for the working capital to total assets, B stands for the net profit before interest and taxes/total assets, C stands for the net profit before taxes to current liabilities ratio, and D represents the total sales to total assets ratio. An S-Score higher than 0.862 indicates that the company is financially healthy. If the value is below 0.862, it indicates that the firm is in a state of crisis'. The independent variables employed in this research are detailed in Table 1.

3.2. Statistical model

Logit regression is a common classification technique for insolvency prediction. It allows for parameter estimation testing and enables researchers to generate probabilities of financial distress for classification purposes (Hernandez Tinoco & Wilson, 2013). Additionally, Ben Jabeur (2017) and Chen et al. (2020) found that the logit model is efficient and provides a high level of classification accuracy. This study uses a comprehensive approach to develop and validate models for predicting the likelihood of financial distress, which includes panel data to ensure robustness and accuracy. This equation provides the log-odds of financial distress based on the given independent variables. The Model 1 as follows:

$$DL_{it} = \text{Constant} + \beta_1 CFO_{it} + \beta_2 CR_{it} + \beta_3 QR_{it} + \beta_4 NWC_{it} + \beta_5 NIR_{it} + \beta_6 IER_{it} + \beta_7 LR_{it} + \beta_8 TLTA_{it} + \beta_9 TAT_{it} + \beta_{10} ROA_{it} + \beta_{11} Size_{it} + \beta_{12} BE_{it} + \beta_{13} GDPG_{it} + \beta_{14} IN_{it} + \beta_{15} UI_{it} + e_{it}.$$

The above equation shows the financial and macroeconomic variables with subscript notation i denoting firm-specific observations and t indicating the time period. Where DL_{it} refers to distress level of firm i at time t , CFO_{it} refers to cash flow from operations, CR_{it} refers to current ratio, QR_{it} refers to quick ratio, NWC_{it} refers to Net Working Capital, NIR_{it} refers to Net Income Ratio, IER_{it} refers to Interest Expense Ratio, LR_{it} refers to Leverage ratio, $TLTA_{it}$ refers to Total Assets to Total Liability ratio, TAT_{it} refers to Total assets turnover ratio, ROA_{it} refers to Return on Assets, $Size_{it}$ refers to Size of the firm, BE_{it} refers to Book value of equity, $GDPG_{it}$ refers to GDP Growth, IN_{it} refers to Inflation, UI_{it} refers to Uncertainty index and e_{it} refers to the error term that captures the influence of unobserved factors and random variability not accounted for by the explanatory variables in the model.

Table 2. Springate model classification.

Status	Company-year observation
Financial distressed	6328
Financial healthy	1129
	7457

Source: Author's computation.

Table 3. Summary of variance inflation factor.

Variables	VIF
CFO	1.012
CR	1.191
QR	1.489
NWC	1.002
NIR	1.016
IER	1.105
LR	1.001
TLTA	1.085
TAT	1.363
ROA	1.101
Size	1.322
BE	1.039
GDPG	1.521
IN	1.344
UI	1.378

Source: Author's computation.

Table 4. Summary of logit regression Model 1.

	Coefficient	p-value
const	-3.95208	<0.0001***
CFO	-8.03666e-07	0.8089
CR	0.57477	<0.0001***
QR	0.45354	<0.0001***
NWC	9.64E-07	0.0152**
NIR	-0.0120189	0.0002***
IER	0.01086	0.0002***
LR	0.04921	<0.0001***
TLTA	-0.000116959	0.0002***
TAT	4.30467	<0.0001***
ROA	6.47E-05	0.3756
Size	-0.341991	<0.0001***
BE	0.28977	<0.0001***
GDPG	-0.105783	<0.0001***
IN	0.05648	<0.0001***
UI	-0.691731	<0.0001***

Source: Author's computation. Note. ***p <.01, **p <.05, *p <.1.

4. Results and discussion

This section presents the results of our logit regression analysis, examining the financial distress of small companies using financial ratios, firm-specific variables, and macroeconomic factors. The discussion interprets the significance and implications of these findings within the context of existing literature and practical applications.

Table 2 refers that the Springate Model was used to classify the firms based on their financial status. According to the model's results, 6328 company-year observations are classified as financially distressed, and 1,129 company-year observations are classified as a financially healthy. Table 3 represents the VIF values for all variables are that below 1.6, suggesting no severe multicollinearity among the predictors.

Table 4 represents Model 1, the logistic regression model's baseline log-odds of financial distress without any predictors, which are significantly below zero, with an intercept of -3.95208 ($p < 0.0001$), indicating a low base probability of financial distress. Several predictors uniquely influence the likelihood of financial distress that are similar to and contrast with the previous studies:

Cash flow from operations does not significantly affect financial distress (-8.03666e-07, $p = 0.8089$), highlighting the complex nature of cash flow impacts. Hence, we reject the H1 and conclude that there is no significant negative relationship between cash flow from operations and the likelihood of financial distress in firms, which is contrary to the theory (Hashi, 1997).

An elevated current ratio significantly increases the likelihood of financial distress (0.574773, $p < 0.0001$), which contrasts with the previous findings (Chen et al., 2020; Waqas & Md-Rus, 2018). This counterintuitive finding suggests that companies with higher current assets relative to current liabilities might not be utilizing their assets efficiently, potentially masking underlying liquidity issues. Similarly, an elevated quick ratio increases the likelihood of financial distress (0.453544, $p < 0.0001$), similar to the current ratio, indicating inefficiencies in asset management. Our results contradict those of Chen et al. (2020; Mselmi et al., 2017). Likewise, increased net working capital slightly raises the likelihood of financial distress (9.64E-07, $p = 0.0152$), reflecting inefficient use of working capital in distressed firms' management. This finding, too, diverges from the results reported by Chen et al. (2020; Mselmi et al., 2017). Our findings indicate a positive association between liquidity ratios and financial distress, which does not support H2.

A higher net income ratio slightly decreases the likelihood of financial distress (-0.0120189, $p = 0.0002$), indicating that higher profitability can mitigate distress risks that align with the findings of Chen et al. (2020; Waqas & Md-Rus, 2018). Thus, our findings support H3.

The interest expense ratio positively impacts the likelihood of financial distress (0.0108575, $p = 0.0002$), suggesting that higher interest expenses due to greater debt burdens can increase the risk of financial distress. These results are consistent with the findings of Chen et al. (2020), thereby validating H4.

Solvency ratios: The Total Liabilities to Total Assets (TLTA) Ratio slightly decreases the likelihood of financial distress (-0.000116959, $p = 0.0002$), suggesting that companies with higher leverage might manage their liabilities better than expected, possibly due prudent financial strategies. These results align

with the research of Bauer and Agarwal (2014), supporting H5. However, the leverage ratio shows a positive association that contradicts the findings of Bauer and Agarwal (2014), suggesting that the hypothesis is not fully supported by the data.

Similarly to Chen et al. (2020), a higher total assets turnover ratio drastically increases the likelihood of financial distress (4.30467, $p < 0.0001$), indicating that firms aggressively utilizing their assets to generate sales might be over-leveraging, leading to financial instability (Alifiah, 2014). Therefore, our results support H6.

The ROA has no significant effect on financial distress (6.47E-05, $p = 0.3756$), suggesting that profitability alone is not a decisive factor. Hence, we reject the H7 and conclude that there is no significant negative relationship between ROA and the likelihood of financial distress in firms, which is inconsistent with the findings of Dewi and Wahyuliana (2019).

Financial distress is less common among larger firms (-0.341991, $p < 0.0001$), likely due to diversified operations and better access to capital, which is consistent with the findings of Rico et al. (2021). Thus, our results affirm this H8. Higher book value of equity increases the likelihood of financial distress (0.28977, $p < 0.0001$), potentially indicating that distressed firms might be overvalued relative to their actual performance as suggested by (Barth et al., 1998; Davis-Friday et al., 2006). Consequently, our findings also support this H9.

Higher GDP growth reduces the likelihood of financial distress (-0.105783, $p < 0.0001$), as economic expansions typically improve business conditions, reducing financial pressures on firms, as proved by Li et al. (2021). Hence, our findings support this H10. Conversely, higher inflation increases the likelihood of financial distress (0.0564846, $p < 0.0001$), likely due to rising costs outpacing revenues, which is consistent with the findings of Pham Vo Ninh et al. (2018). Thus, our findings validate this H11 as well. A higher Uncertainty Index reduces the likelihood of financial distress (-0.691731, $p < 0.0001$), which aligns with the findings of Li et al. (2021) and Phan et al. (2021), suggesting that firms' adaptive strategies during uncertain periods enhance their resilience. Consequently, our results support this H12. Table 5 shows that the Model 1 accuracy is 92.7%, and it is statistically significant with a Chi-square value of 3503.82 ($p < 0.0001$).

Table 6 represents a robustness check. To ensure robustness, we conducted a second logistic regression using a subset of the observations (Model 2). The results align with Model 1, reaffirming the significance and direction of the key predictors. Without any predictors, the baseline log-odds for financial

Table 5. Summary of Model 1 accuracy.

Observed	Predicted	
	Distressed (company-year observations)	Healthy (company-year observations)
Distressed	6195	133
Healthy	409	720

Source: Author's computation.

Table 6. Summary of logit regression Model 2.

	Coefficient	p-value
const	-3.43923	0.0009***
CFO	4.87E-06	0.687
CR	0.48688	<0.0001***
QR	0.56656	<0.0001***
NWC	1.20E-05	0.0027***
NIR	-0.0226147	<0.0001***
IER	0.02389	<0.0001***
LR	-0.529112	0.1212
TLTA	-0.000127420	0.0022***
TAT	4.17924	<0.0001***
ROA	-3.41686e-05	0.6816
Size	-0.333783	0.0002***
BE	0.2609	<0.0001***
GDPG	-0.0874433	0.0011***
IN	0.06649	0.0007***
UI	-0.767978	0.0001***

Source: Author's computation. Note. *** $p < .01$, ** $p < .05$, * $p < .1$.

Table 7. Summary of Model 2 accuracy.

Observed	Predicted	
	Distressed (company-year observations)	Healthy (company-year observations)
Distressed	2427	63
Healthy	162	312

Source: Author's computation.

Table 8. Key comparisons between Models 1 & 2.

Variables	Model 1 coefficient	Model 2 coefficient	Model 1 p-value	Model 2 p-value
CR	0.574773	0.486883	<0.0001***	<0.0001***
QR	0.453544	0.566561	<0.0001***	<0.0001***
NWC	9.64E-07	1.20E-05	0.0152**	0.0027***
NIR	-0.0120189	-0.0226147	0.0002***	<0.0001***
IER	0.0108575	0.0238896	0.0002***	<0.0001***
TLTA	-0.00011695	-0.00012742	0.0002***	0.0022***
TAT	4.30467	4.17924	<0.0001***	<0.0001***
Size	-0.341991	-0.333783	<0.0001***	0.0002***
BE	0.28977	0.260895	<0.0001***	<0.0001***
GDPG	-0.105783	-0.0874433	<0.0001***	0.0011***
IN	0.0564846	0.0664876	<0.0001***	0.0007***
UI	-0.691731	-0.767978	<0.0001***	0.0001***

Source: Author's computation. Note. ***p < .01, **p < .05, *p < .1.

distress are -3.43923 ($p = 0.0009$). Key predictors such as the interest expense ratio (0.0238896, $p < 0.0001$), current ratio (0.486883, $p < 0.0001$), and total asset turnover (4.17924, $p < 0.0001$) remain significant, consistent with Model 1. Other predictors like the TLTA Ratio (-0.000127420, $p = 0.0022$), Net Working Capital (1.20E-05, $p = 0.0027$), and Quick Ratio (0.566561, $p < 0.0001$) also show similar significance and direction. The size of the firm (-0.333783, $p = 0.0002$), book value of equity (0.260895, $p < 0.0001$), GDP growth (-0.0874433, $p = 0.0011$), inflation (0.0664876, $p = 0.0007$), and uncertainty index (-0.767978, $p = 0.0001$) follow the same trends as in Model 1.

Table 7 includes the performance metrics for Model 2, which is also robust, with a McFadden R-squared of 0.544228, explaining around 54% of the variance. The model's accuracy is 92.4%, and it is statistically significant with a Chi-square value of 1418.02 ($p < 0.0001$). Table 8 represents that the consistency in key predictors and performance metrics between the two models provides strong evidence of robustness. This strengthens the reliability of our analysis and the validity of our conclusions.

5. Conclusion

This study provides insights into financial distress among small companies in emerging markets like India. While researchers have predominantly focused on predicting the financial distress of large-scale companies, small-scale businesses have received comparatively less attention in previous studies. The study combines financial, firm, and macroeconomic variables to construct a robust model for predicting financial health and identify key factors contributing to the financial distress. All hypotheses were supported by prior studies, providing significant evidence and confirming the proposed relationship, except for H1, H2, and H7.

The findings highlight the significance of variables such as the interest expense ratio, current ratio, total assets turnover ratio, TLTA ratio, net working capital, quick ratio, net income ratio, size of the company, book value of equity, GDP growth, inflation, and economic policy uncertainty in predicting the financial health of small companies. The predictive models developed in this study demonstrated high accuracy, with initial Model 1 at 92.47% accuracy and Model 2 at 92.4%. These results underscore the potential of the models to serve as practical tools for stakeholders to enable them to proactively address financial vulnerabilities in small enterprises.

The findings also emphasize the complexity of maintaining financial health for small businesses and underscore the need for tailored financial strategies that address their unique challenges. The models' ability to effectively predict financial distress can help business owners, financial institutions, and policy-makers take preventive actions to mitigate potential risks (Jayasekera, 2018).

Furthermore, the study highlights the pronounced impact of macroeconomic conditions on the stability of a firm. This underscores the importance of sustaining macroeconomic stability and developing targeted financial policies for small enterprises. By maintaining stable economic conditions, policy makers can help prevent the economy from falling into distress, thereby mitigating the adverse effects of volatility on small businesses.

In addition to their high academic value, the findings of this research are highly relevant for practice. For example, predictive models can help financial institutions in determining the creditworthiness of small businesses with better accuracy. Policymakers could use these insights to better target support for small businesses in financial distress. Additionally, small business operators can benefit from understanding the key financial ratios and macroeconomic indicators that influence their firms financial stability (Phan et al., 2021).

This study also offers valuable insights into how financial distress evolves in small firms. These findings are useful for practical decisions and help small businesses become more resilient in times of economic insecurity. By reinforcing the significance of firm-specific financial soundness and broader economic conditions, the study offers a comprehensive view of financial stability for small enterprises.

Future research could extend these results by conducting longitudinal research that follows small businesses across time in order to offer insight into processes of financial distress. Additionally, more variables could augment prediction as well as generalizability in future models. For example, the practice of handling variables related to market conditions, firm-specific management practices, and industry-level factors could reveal deeper causes and predictors of financial distress.

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Authors' contributions

CRedit: **Rampilla Mahesh:** Data curation, Formal analysis, Methodology, Software; **Anamika Kumari:** Conceptualization; **Debabrata Sharma:** Data curation, Supervision; **Sreelakshmi N:** Methodology; **Ranjana Ramesh:** Methodology.

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Data availability statement

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

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